

MONTROSE 11 ENE, CO, USA

WMO: 745220

Lat: 38.5440N

Lon: 107.6928W

Elev: 8402

StdP: 10.75

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 03060

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.6	(c) 5.0	(d) -7.3	(e) 5.1	(f) 10.2	(g) -3.7	(h) 6.2	(i) 13.5	(j) 13.1	(k) 30.6	(l) 11.0	(m) 27.5	(n) 3.0	(o) N/A	(p) 0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 23.0	(c) 80.8	(d) 52.1	(e) 78.6	(f) 51.8	(g) 76.5	(h) 51.4	(i) 56.9	(j) 70.5	(k) 55.7	(l) 70.2	(m) 54.6	(n) 69.4	(o) 5.3	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
52.8	81.7	59.1	51.0	76.2	58.9	49.4	71.8	58.3	28.2	70.7	27.3	70.3	26.5	69.8	61.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	9.6	8.3	DB	-5.9	84.6	4.7	1.5	-9.3	85.7	-12.0	86.5	-14.7	87.4	-18.1	88.5
			WB	-5.9	58.7	5.8	1.5	-10.0	59.8	-13.4	60.6	-16.7	61.5	-20.9	62.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.9	22.8	25.3	32.8	39.6	48.4	60.2	65.4	62.8	56.0	43.5	33.5	23.3		
	DBStd	16.78	7.73	8.25	8.25	8.26	7.93	6.39	3.66	3.84	6.61	8.46	8.97	9.01		
	HDD50	4100	843	693	533	319	126	9	0	0	27	225	496	829		
	HDD65	8164	1308	1113	998	762	515	160	40	88	274	666	946	1294		
	CDD50	1502	0	0	0	7	76	316	477	396	206	24	0	0		
	CDD65	90	0	0	0	0	0	17	51	19	3	0	0	0		
	CDH74	1045	0	0	0	0	3	239	508	247	48	0	0	0		
	CDH80	87	0	0	0	0	0	18	54	12	3	0	0	0		
(14) Wind	WSAvg	4.0	3.8	4.1	4.4	4.7	4.4	4.5	3.8	3.7	3.9	3.7	3.6	3.6		
(15) Precipitation	PrecAvg	14.0	1.1	1.1	1.2	1.3	1.1	0.6	1.1	1.3	1.6	1.5	1.1	1.1		
	PrecMax	18.4	2.3	2.6	2.9	3.7	3.1	1.6	2.4	2.1	3.0	4.4	1.9	2.9		
	PrecMin	10.0	0.2	0.2	0.0	0.5	0.0	0.0	0.1	0.4	0.4	0.3	0.3	0.6		
	PrecStd	2.2	0.5	0.6	0.6	0.7	0.9	0.4	0.7	0.5	0.8	0.9	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	48.4	59.4	65.3	73.4	82.8	84.1	81.8	78.9	68.1	58.4	47.0		
		MCWB	32.5	35.0	40.0	42.5	46.4	51.9	53.6	51.6	51.6	45.2	39.6	33.4		
	2%	DB	40.1	44.1	54.4	61.1	69.2	79.2	81.6	79.3	74.9	63.9	54.1	42.1		
		MCWB	30.1	32.8	37.6	40.7	45.1	50.8	53.0	51.4	49.6	43.6	38.3	31.2		
	5%	DB	36.6	41.0	49.7	57.7	66.4	76.9	79.3	77.0	72.0	60.6	50.6	38.9		
		MCWB	28.6	31.1	35.3	38.9	43.9	49.8	53.1	51.5	48.4	42.3	36.3	30.2		
	10%	DB	33.9	37.3	46.0	54.2	63.4	74.4	77.0	74.5	69.2	57.1	46.9	36.0		
		MCWB	27.4	29.1	33.3	37.8	42.8	48.8	52.8	51.5	47.3	40.7	34.3	29.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.8	36.5	40.9	43.8	48.3	55.7	58.8	57.9	55.0	48.7	41.3	35.3		
		MCDB	41.2	46.2	58.0	61.7	66.5	75.2	71.8	68.4	70.3	59.5	53.8	42.6		
	2%	WB	32.2	34.1	38.6	41.8	46.2	53.6	57.3	56.4	53.1	46.4	39.0	33.5		
		MCDB	37.7	43.0	53.6	57.6	64.5	73.7	71.1	69.1	67.4	59.3	51.4	39.3		
	5%	WB	30.4	32.4	36.3	40.3	44.7	52.2	56.3	55.2	51.6	44.3	36.8	31.8		
		MCDB	35.7	40.3	49.4	55.1	62.9	72.3	70.5	69.3	65.5	57.6	48.4	37.8		
	10%	WB	28.4	30.9	34.4	38.5	43.4	50.6	55.3	54.1	50.2	42.6	34.8	30.0		
		MCDB	33.6	38.0	45.8	52.3	61.2	70.5	69.8	68.4	64.0	55.2	45.3	36.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	15.2	17.5	19.9	21.0	23.5	23.0	22.4	22.1	19.4	17.9	14.6		
		MCDBR	19.1	18.7	21.2	24.5	24.7	25.9	25.5	25.6	25.3	23.7	22.0	16.6		
		MCWBR	12.8	10.9	10.4	11.6	10.9	10.6	8.9	8.9	9.7	10.9	11.4	9.9		
	5% WB	MCDBR	16.7	16.3	20.5	22.3	23.8	23.7	21.3	21.6	21.1	20.6	20.6	14.7		
		MCWBR	11.4	9.7	10.2	10.9	10.8	9.9	8.0	8.3	8.3	9.7	11.0	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.216	0.220	0.245	0.264	0.276	0.275	0.312	0.308	0.271	0.249	0.225	0.217		
	taud		2.636	2.651	2.595	2.570	2.589	2.592	2.516	2.519	2.650	2.682	2.685	2.625		
	Ebn at Noon		320	330	327	323	318	317	304	304	313	314	314	312		
	Edh at Noon		21	24	28	31	31	31	33	32	26	23	20	20		
(44) All-Sky Solar Radiation	RadAvg		830	1083	1531	1872	2137	2478	2189	1922	1696	1274	912	708		
	RadStd		53	77	125	115	167	157	126	88	94	104	68	57		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page